

* SVM - Theory

- The SVM first finds a separating hyperplane ~~find line~~
- then it creates the two margin planes ~~the~~, one on each side of the hyperplane
- It extends these margin planes outward until they 'touch' the closest data points from each class
- those points that lie exactly on the marginal planes are called as **support vectors**

* Distance from a point to an SVM hyperplane

- The standard equation for a hyperplane in SVM is:

$$w^T x + b = 0$$

The distance from any point x to the hyperplane is:

$$\text{distance} = \frac{|w^T x + b|}{\|w\|}$$

- w = weight vector (coefficients)
- x = point you're measuring from
- b = intercept (bias)
- $\|w\|$ = magnitude (length) of w :

$$\sqrt{w_1^2 + w_2^2 + w_3^2 + \dots + w_n^2}$$

- Using "+", "-", and "0"

In SVM, we define

- Decision boundary (hyperplane): $w^T x + b = 0$
- Positive margin: $w^T x + b = +1$ (the nearest positive-class points)
- Negative margin: $w^T x + b = -1$ (the nearest negative-class points)

These "+1" and "-1" lines/planes are margins, set this way so that the closest points (support vectors) from each class will sit exactly on them.

- Calculating Margin Width Step by step

Distance from hyperplane ($=0$) to positive margin ($+1$) plane:

$$d_+ = \frac{|1 - 0|}{\|w\|} = \frac{1}{\|w\|}$$

Distance from hyperplane ($=0$) to negative margin (-1) plane:

$$d_- = \frac{|-1 - 0|}{\|w\|} = \frac{1}{\|w\|}$$

Total margin width (distance between $+1$ and -1 planes)

$$d_+ + d_- = \frac{1}{\|w\|} + \frac{1}{\|w\|} = \frac{2}{\|w\|}$$

⇒ This is what SVM tries to maximize

$$d_- = \frac{|-1 - 0|}{\|w\|} = -1 \quad (\text{But the formula wraps this in absolute value: } |-1| = 1)$$

So, the result is always positive (since distance can't be negative)

★ How do you maximize the margin?

→ You want to make $\|w\|$ as small as possible

→ Making $\|w\|$ smaller makes the denominator smaller, which makes $\frac{2}{\|w\|}$ larger (i.e. the margin increases)

→ In optimization terms:

• SVM minimizes $\frac{1}{2} \|w\|^2$ (minimizing the squared norm makes $\|w\|$ small)

• This automatically leads to maximizing the margin

Maximizing - $\|w\|$

1. The Margin Formula

$$\text{Margin} = \frac{2}{\|w\|}$$

Here, $\|w\|$ means "the length (magnitude of the weight vector)"

2. What does "Inverse" mean here?

→ "Inverse" here means as one gets bigger, the other gets smaller, and vice versa:

• If $\|w\|$ is small, then $\frac{2}{\|w\|}$ is large ("margin increases")

• If $\|w\|$ is large, then $\frac{2}{\|w\|}$ is small ("margin shrinks")

For eg:

→ If $\|w\| = 1$, $\text{margin} = \frac{2}{\|w\|} = \frac{2}{1} = 2$

→ If $\|w\| = 2$, $\text{margin} = \frac{2}{2} = 1$ (margin got smaller.)

3. Why do we minimize $\|w\|$ to maximize the margin?

• The mathematical relationship is $\text{margin} = 2$ divide by $\|w\|$

• To make the margin as large as possible, you want the denominator ($\|w\|$) as small as possible (but > 0)

• This is exactly why SVM's objective is to find the smallest possible value of $\|w\|$ that still separates the classes

4. Simple Analogy

Think of:

• $\text{Speed} = \frac{\text{Distance}}{\text{Time}}$ ∴ To maximize speed for a fixed distance you minimize the time

• Similarly, $\text{Margin} = \frac{2}{\|w\|}$ ∴ To maximize margin, you minimize $\|w\|$

- In Math we don't minimize $\|w\|$
- Instead we convert it like this $\frac{1}{2} \|w\|^2$ in order to minimize

But why? not $\|w\|$ directly instead $\frac{1}{2} \|w\|^2$

⇒ In optimization (calculus / convex optimization) it's easier to minimize a squared term than the square root:

$$\|w\| = \sqrt{w_1^2 + w_2^2 + w_3^2 + \dots + w_n^2}$$

↳ The square root makes derivatives messy

If we square it, it becomes:

$$\|w\|^2 = w_1^2 + w_2^2 + w_3^2 + \dots + w_n^2$$

Now the derivative is simple: derivative of w_i^2 is $2w_i \rightarrow$ a bit messy.

Multiply by $\frac{1}{2} \left(\frac{1}{2} w_i^2 \right) \rightarrow$ derivative becomes just w_i

⇒ So, the $1/2$ is only for convenience, to make derivatives simpler

It does not change the optimization goal

5) $\rightarrow \|w\| > 0$: Margin is "finite" and positive - this is valid for SVM

$\rightarrow \|w\| = 0$: Margin is "infinite", but this is not possible for a functional SVM model

$\rightarrow$ (hyperplane becomes meaningless) :- All points are equally separated, or the weights have no direction)

$\rightarrow$ so SVM never allows $\|w\| = 0$

$\rightarrow \|w\| < 0$: Not possible for vector norms; always positive or zero

CF :- Minimize $\frac{\|w\|}{2}$
(w, b)

* For minimizing equation :-

$$\Rightarrow \frac{1}{2} \|w\|^2 \text{ or } \|w\| \text{ or } \frac{\|w\|}{2}$$

$$w, b : \frac{1}{2} \|w\|^2$$

* Cost or Loss Function :

$$\Rightarrow \frac{1}{2} \|w\|^2 \text{ or equivalently, } \frac{\|w\|}{2}$$

serves as the cost function (objective function) in SVM

* Soft Margin SVM Cost Function or SVM Primal Objective Function with slack Variables

$$\Rightarrow \text{Cost} = \frac{1}{2} \|w\|^2 + c \sum_{i=1}^n \xi_i$$

- The first term ($\frac{1}{2} \|w\|^2$) - tries to maximize the margin
- The second term ($c \sum_{i=1}^n \xi_i$) - penalizes misclassification and margin violations, where:
 - ξ_i are slack variables that allow some points to be on the wrong side of the margin / hyperplane
 - c is regularization parameter that controls the balance / tradeoff between having a wide margin and following allowing misclassifications (margin violations)

6) Hyperplane & Marginal plane

- When maximizing the margin, the data points stay fixed in their original positions; only the marginal planes and the hyperplane move to maximize the separation.
- If any data points overlap with the margins to get misclassified, it is because the boundary moved, not because the data points moved. The data itself never changes position during SVM ~~machine~~ training.

7) Why we maximize marginal (distance)

- If the margin (distance) is maximized:
 - The model is more robust and generalizes better on new or unseen data
 - It is less sensitive to noise and outliers
 - Classification is more reliable for unseen points
- If the margin (distance) is small:
 - The model is ~~more~~ more likely to overfit and become sensitive to minor variations or noise
 - It may misclassify new data more easily
 - Generalization to new data is ~~useless~~ ^{poor}

Less sensitive to noise & outliers means

⇒ Model does not change its predictions or decision boundary easily if there are a few unusual, incorrect or extreme data points, it ignores or downplays these minor or rare problems, making predictions more stable and reliable.

⇒ noise in data :- age in name field or puts a name where the age should be this is an example of noise in data.